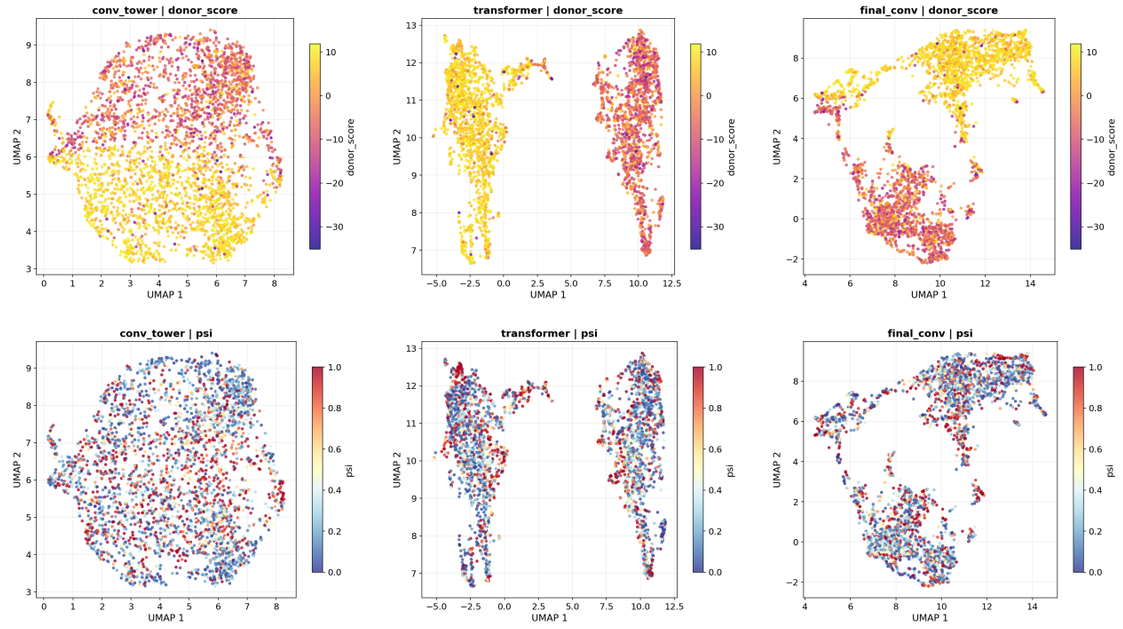
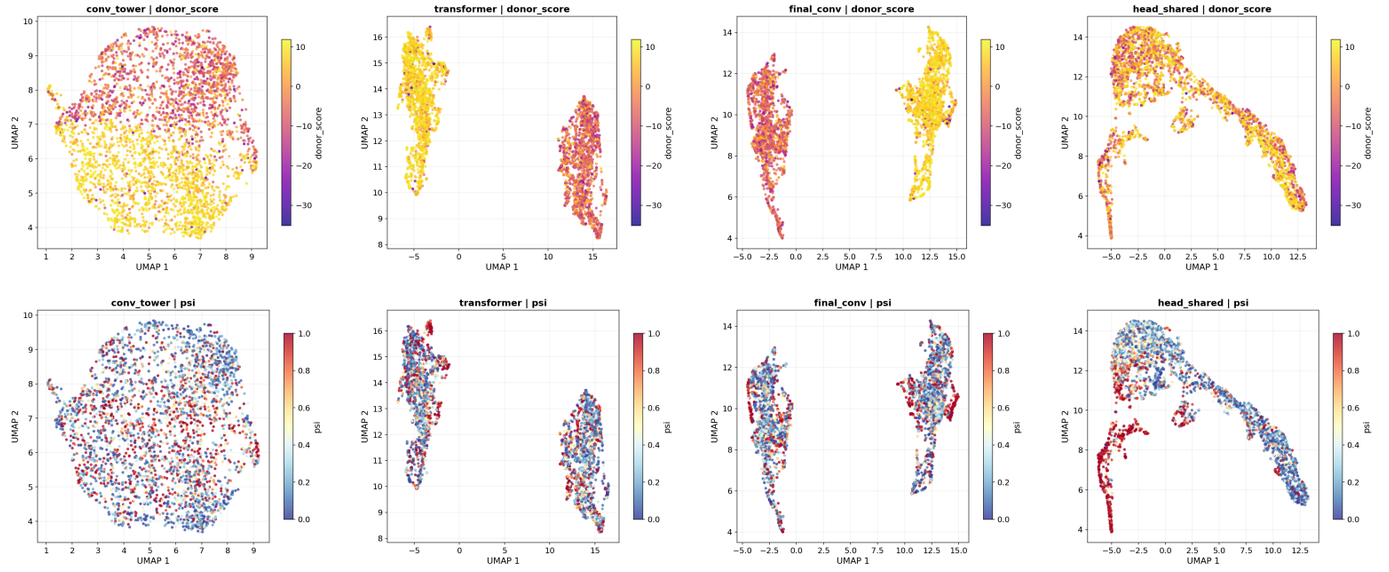
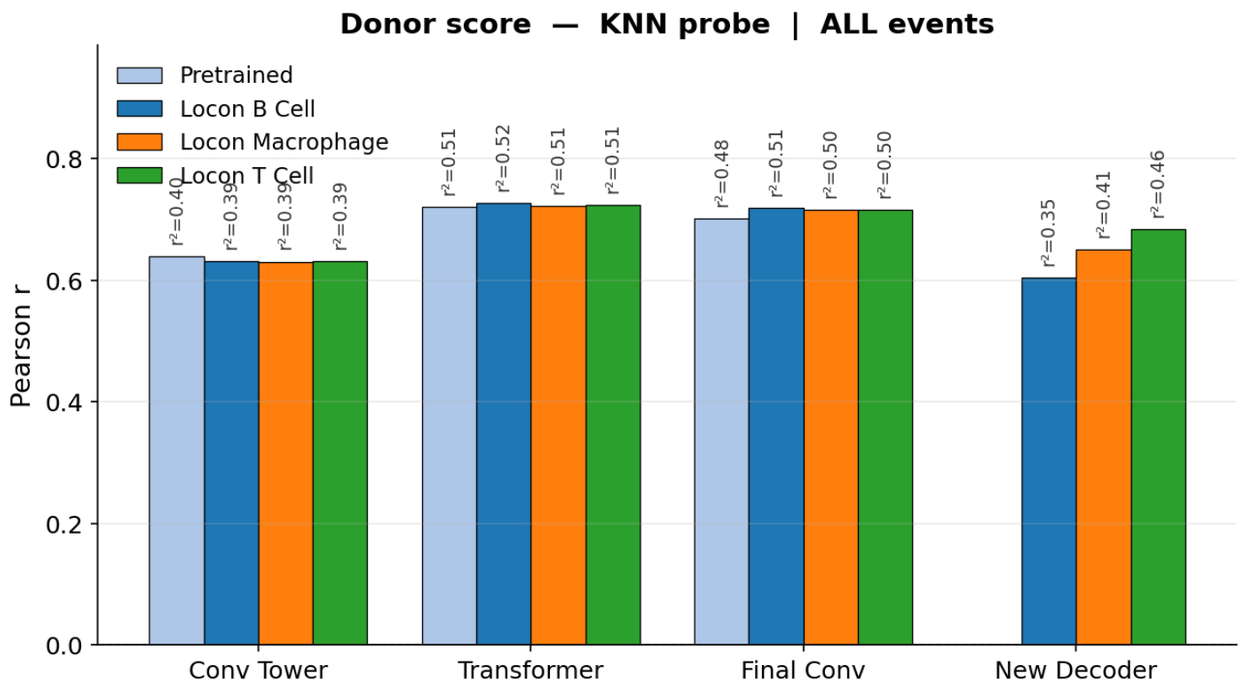
A


conv_tower | donor_score
transformer | donor_score
final_conv | donor_score
conv_tower | psi
transformer | psi
final_conv | psi
UMAP 1
UMAP 2
donor_score
psi

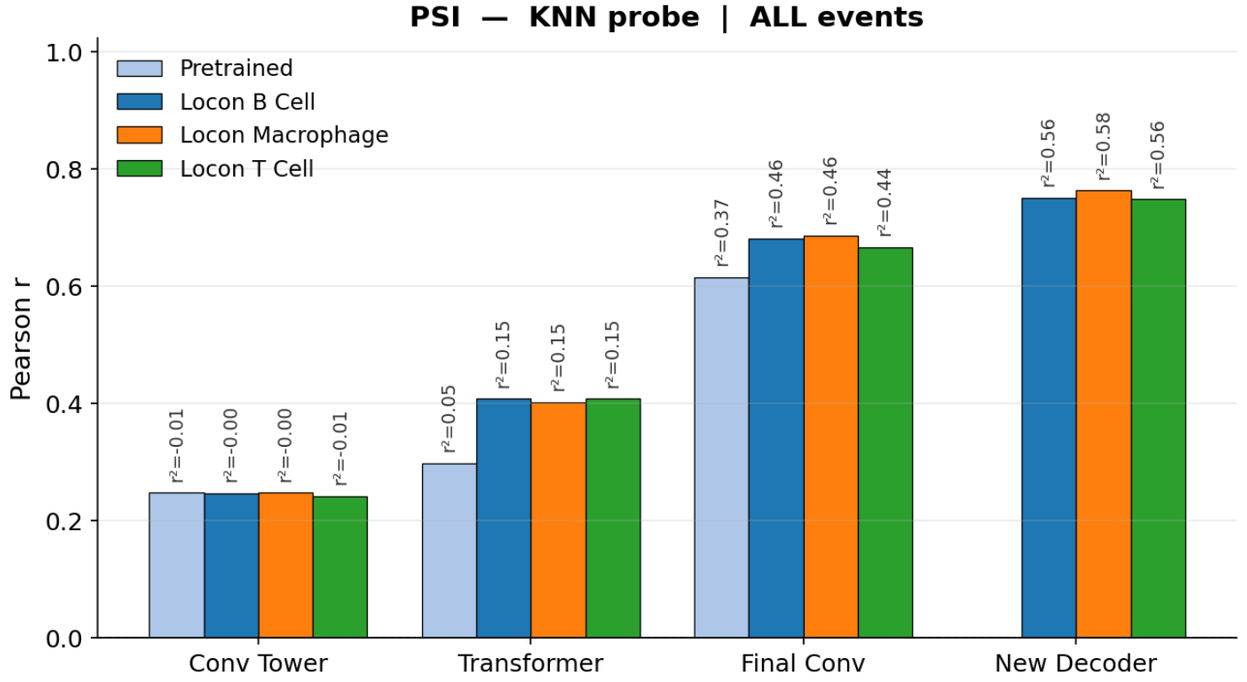
B


conv_tower | donor_score
transformer | donor_score
final_conv | donor_score
head_shared | donor_score
conv_tower | psi
transformer | psi
final_conv | psi
head_shared | psi
UMAP 1
UMAP 2
donor_score
psi

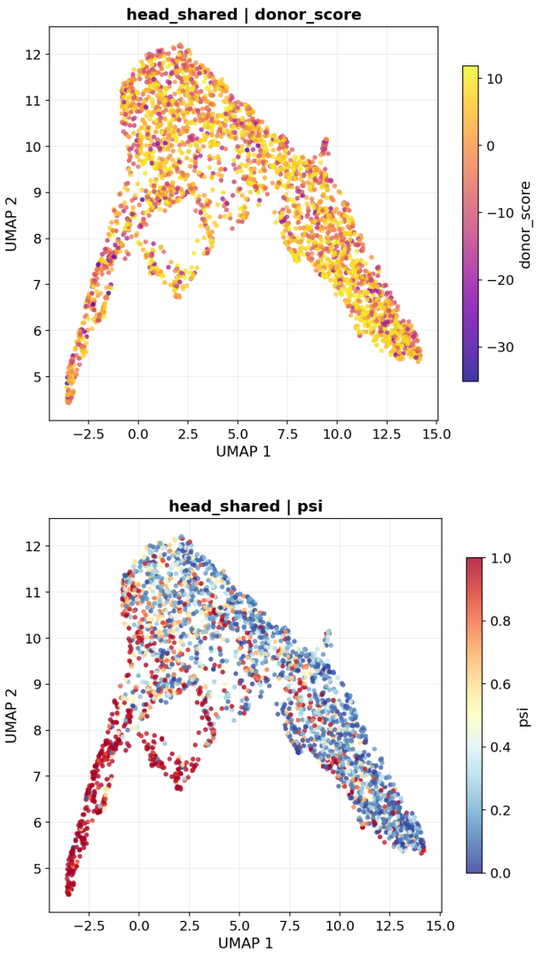
C


Donor score — KNN probe | ALL events
Pretrained
Locon B Cell
Locon Macrophage
Locon T Cell
Pearson r
r²=0.40
r²=0.39
r²=0.39
r²=0.39
r²=0.51
r²=0.52
r²=0.51
r²=0.51
r²=0.48
r²=0.51
r²=0.50
r²=0.50
r²=0.35
r²=0.41
r²=0.46
Conv Tower
Transformer
Final Conv
New Decoder

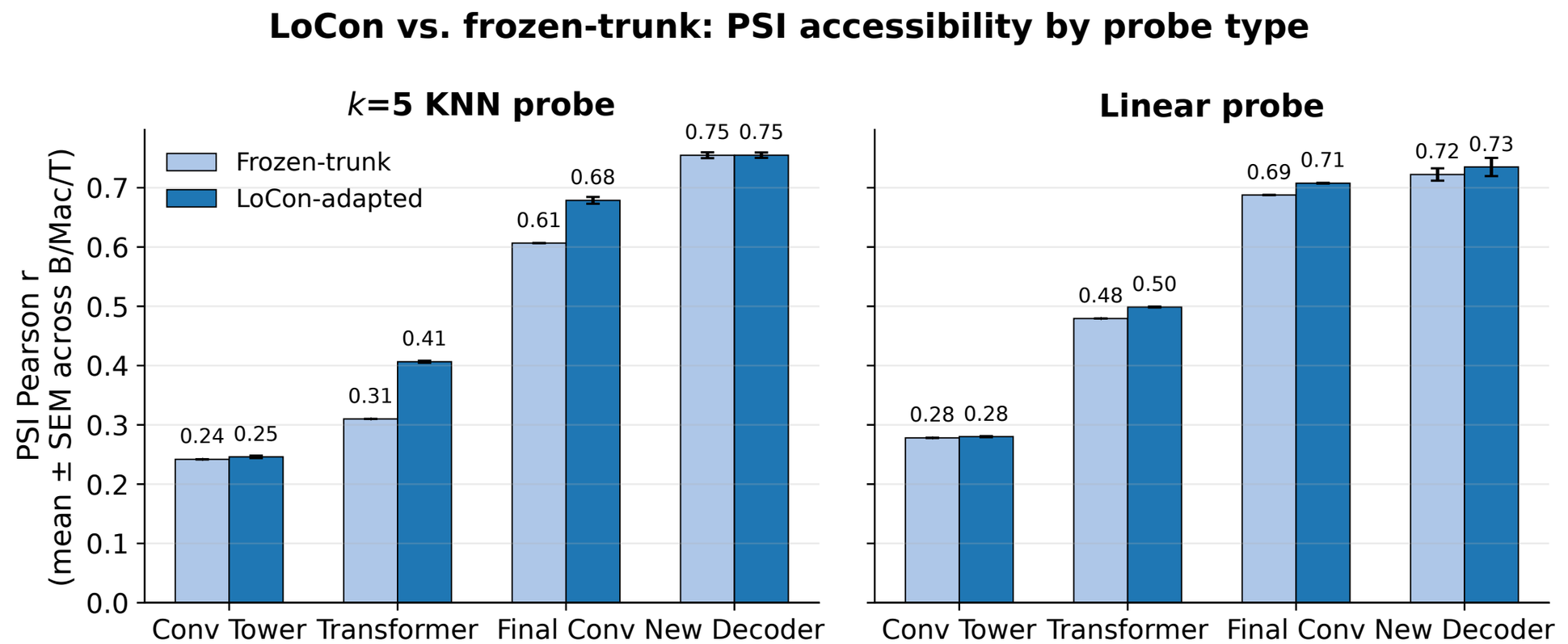
D


PSI — KNN probe | ALL events
Pretrained
Locon B Cell
Locon Macrophage
Locon T Cell
Pearson r
r²=-0.01
r²=-0.00
r²=-0.00
r²=-0.01
r²=0.05
r²=0.15
r²=0.15
r²=0.15
r²=0.37
r²=0.46
r²=0.46
r²=0.44
r²=0.56
r²=0.58
r²=0.56
Conv Tower
Transformer
Final Conv
New Decoder

E


head_shared | donor_score
head_shared | psi
UMAP 1
UMAP 2
donor_score
psi

F


LoCon vs. frozen-trunk: PSI accessibility by probe type
k=5 KNN probe
Linear probe
Frozen-trunk
LoCon-adapted
PSI Pearson r
(mean ± SEM across B/Mac/T)
0.24 0.25
0.31
0.41
0.61
0.68
0.75 0.75
0.28 0.28
0.48 0.50
0.69 0.71
0.72 0.73
Conv Tower
Transformer
Final Conv
New Decoder